The long-term effect of indicating AI fallibility on automation bias in mental health application users.

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Figure 1: Seattle Mariners at Spring Training, 2010.

Abstract

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CCS Concepts

• **Information systems** → **Users and interactive retrieval**; **Evaluation of retrieval results**; • **General and reference** → **Surveys and overviews**; • **Computing methodologies** → Knowledge representation and reasoning; **Artificial intelligence**; • **Social and professional topics** → Age; Gender; • **Human-centered computing** → Human computer interaction (HCI); **Empirical studies in interaction design**; *Visualization*; **Empirical studies in HCI**.

Keywords

Mental Health, AI, Artificial Intelligence, FWS, STUD, Automation Bias, Fallibility, Mental Health Application, AI Assistant, Long-Term, Long-Term Effects, Priming, Trust, Confidence

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Contents

1 Introduction

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The Latex typesetting markup language is specially suitable for documents that include mathematics.

2 Definitions

2.1 Participants

We define the following sets:

$$\begin{aligned}\text{Groups } G &:= \{\text{Active}, \text{Control}\},\\ \text{Case Vignettes } C &:= \{\text{Amina}, \text{Aylin}, \text{Lena}, \text{Martina}, \text{Mathias}\},\\ \text{Truthfulness } T &:= \{\text{Truth}, \text{Untruth}\}.\end{aligned}$$

A *round* is a pair consisting of a case vignette and a truthfulness label. Hence, the set of all rounds is

$$R := C \times T.$$

A *session* consists of exactly three rounds. Formally, a session is an ordered triple

$$s = (r_1, r_2, r_3) \in R^3.$$

However, not every element of R^3 is considered valid. A valid session must satisfy the following constraints:

$$|\{i \in \{1,2,3\} \mid r_i = (c, \text{Truth}) \text{ for some } c \in C\}| = 1,$$

i.e. exactly one round in the session is labelled Truth.

Furthermore,

$$\forall i, j \in \{1,2,3\} : \quad i \neq j \Rightarrow c_i \neq c_j,$$

where $r_i = (c_i, t_i)$. That is, no case vignette may occur more than once within the same session.

We denote the set of all valid sessions by

$$S \subseteq R^3.$$

A participant is a triple

$$p = (g, s_1, s_2) \in G \times S \times S,$$

where

$$g \in G, \qquad s_1, s_2 \in S.$$

The two sessions of a participant must share exactly one case vignette. Formally,

$$\left|\left\{c \in C \mid \exists\, t_1, t_2 \in T : (c, t_1) \in \{r_1, r_2, r_3\} \wedge (c, t_2) \in \{r'_1, r'_2, r'_3\}\right\}\right| = 1,$$

where

$$s_1 = (r_1, r_2, r_3), \qquad s_2 = (r'_1, r'_2, r'_3).$$

An example of a valid participant is

$$\begin{aligned}p = (&\text{Active},\\ &((\text{Amina}, \text{Truth}), (\text{Lena}, \text{Untruth}), (\text{Mathias}, \text{Untruth})),\\ &((\text{Martina}, \text{Truth}), (\text{Aylin}, \text{Untruth}), (\text{Mathias}, \text{Untruth}))).\end{aligned}$$

2.2 Participant Allocation

To generate participants satisfying the constraints defined in the previous subsection, we employ a deterministic pseudo-random allocation algorithm, denoted by A_s (Participant Allocator).

For a fixed pseudo-random seed s, the algorithm defines a mapping

$$A_s : \mathbb{N}_0 \to P,$$

where

$$A_s(n) = p_n,$$

with

- P denoting the set of valid participants defined previously,
- $n \in \mathbb{N}_0$ denoting the participant index,
- $p_n \in P$ denoting the generated participant corresponding to index n,
- $s \in \mathbb{N}$ denoting the global pseudo-random seed.

In the present study, the seed value $s = 403$ was used. The algorithm is deterministic with respect to the seed s, thereby ensuring reproducibility of the participant allocation.

The reference Python implementation is provided in the attached file allocate.py.

2.3 Automation bias

Automation bias is modelled as a continuous function that maps evaluation parameters to a scalar value in the interval $[0,1]$, representing the degree of evidence that automation bias has occurred.

The function is defined as:

$$B : \{0,1\} \times \{1,2,3,4,5\} \times [0,100] \to [0,1],$$

$$B(i, r, c),$$

where:

- $B(i, r, c) \in [0,1]$ denotes the automation bias score,
- $i \in \{0,1\}$ indicates correctness of the automated answer (1 = correct, 0 = incorrect),
- $r \in \{1,2,3,4,5\}$ denotes the evaluator's rating of the answer (1 = very bad, 5 = very good),
- $c \in [0,100]$ denotes the evaluator's confidence in the rating.

2.3.1 Definition of the Bias Function. The function is defined as:

$$B(i, r, c) = (1 - i) \cdot \frac{r-1}{5-1} \cdot \frac{c}{100}.$$

2.3.2 Interpretation. The term $(1-i)$ ensures that automation bias is only considered when the automated answer is incorrect. If the answer is correct ($i = 1$), then $B(i, r, c) = 0$ regardless of the other parameters.

The term $\frac{r-1}{4}$ maps the rating to a normalized interval $[0, 1]$, where higher ratings indicate stronger perceived quality of the answer. In the context of automation bias, higher ratings for incorrect answers indicate a stronger tendency toward bias.

The term $\frac{c}{100}$ represents the evaluator's confidence in their rating. Higher confidence strengthens the evidence of automation bias.

By construction, $B(i, r, c) \in [0, 1]$.

2.3.3 Threshold for Minimal Automation Bias. We define a reference threshold based on a minimally biased configuration:

$$(i, r, c) = (0, 3, 50).$$

Substituting into the function yields:

$$\begin{aligned} B(0, 3, 50) &= (1 - 0) \cdot \frac{3 - 1}{4} \cdot \frac{50}{100} \\ &= 1 \cdot \frac{2}{4} \cdot 0.5 \\ &= 1 \cdot 0.5 \cdot 0.5 \\ &= 0.25. \end{aligned}$$

We therefore define the minimal automation bias threshold as:

$$B_{\text{threshold}} = 0.25, \quad \text{and we consider} \quad B(i, r, c) > 0.25$$

as indicating evidence of automation bias.

3 Testing Methods

3.1 General

3.2 Session 1

3.3 Session 2

4 Results

Acknowledgments

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We also thank all testers who participated in the study and made this work possible.

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References

[1] [SW] The CGAL Project, The Computational Geometry Algorithms Library (Coord.by CGAL Editorial Board), 1996. URL: https://cgal.org/.

A Palloc Python Implementation

allocate.py